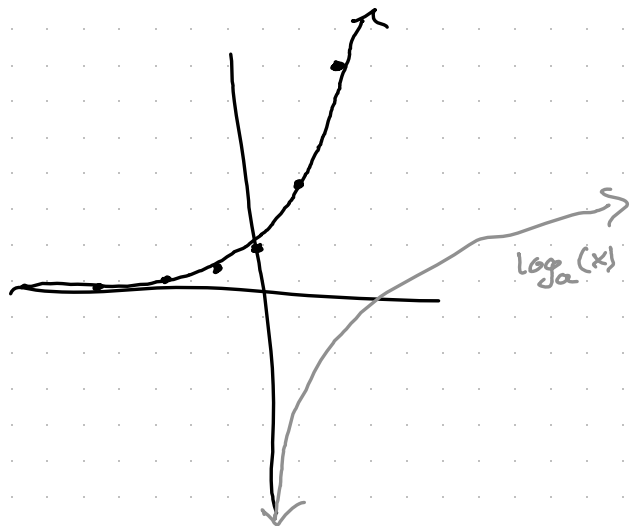


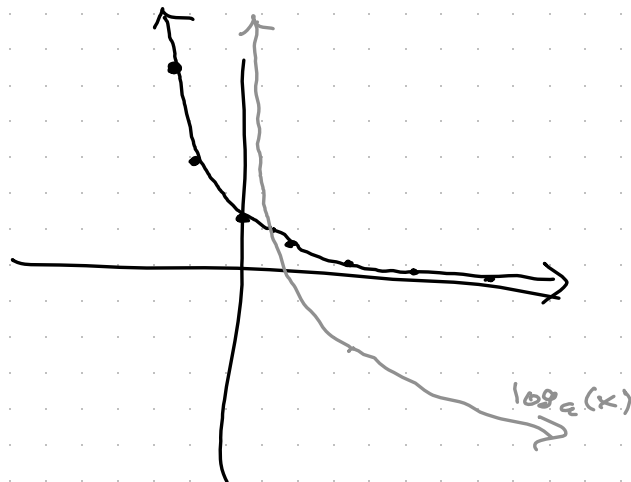
Limits of exponentials and logarithms — 0.1.5



a^x for $a > 1$

$$\lim_{x \rightarrow \infty} a^x = \infty \quad \lim_{x \rightarrow \infty} \log_a(x) = \infty$$

$$\lim_{x \rightarrow -\infty} a^x = 0 \quad \lim_{x \rightarrow 0^+} \log_a(x) = -\infty$$



a^x for $a < 1$

$$\lim_{x \rightarrow \infty} a^x = 0 \quad \lim_{x \rightarrow 0^+} \log_a(x) = \infty$$

$$\lim_{x \rightarrow -\infty} a^x = \infty \quad \lim_{x \rightarrow \infty} \log_a(x) = -\infty$$